
AR 514: Vision and Learning Based Control

Course Code	AR 514
Course Name	Vision and Learning Based Control
L-T-P-C	3-0-0-3
Prerequisites	Consent of faculty advisor
Intended for	UG, PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction:** Overview, motivation, and real-world practical applications. (1 hours)
- **Visual Sensor Model and Calibration:** Camera model, Coordinate Frames and Transforms, Intrinsic camera calibration, and extrinsic camera calibration. (10 hours)
- **Visual Servoing:** Image Jacobian, Robot Jacobian, Image Based Visual Servoing, Position Based Visual Servoing, Eye-in-hand and Eye-to-hand Configurations, Comparison among different class of visual servoing. (14 hours)
- **Robot Learning:** Basic concepts of reinforcement learning, reinforcement learning algorithms. Robot learning by demonstration. (14 hours)
- **Hybrid Method Design:** Comparative analysis for various methods. Explore, understand and identify different ways to design a hybrid scheme to control the given system of interest. Case study and course projects. (3 hours)

Text Books

1. Corke, Peter I., and Oussama Khatib, **Robotics, vision and control: fundamental algorithms in MATLAB**, Vol. 73, Springer, 2011.
2. Vakanski, Aleksandar, and Farrokh Janabi-Sharifi, **Robot learning by visual observation**, John Wiley & Sons, 2017.

References

1. Ijspeert, Auke Jan, et al., Dynamical movement primitives: learning attractor models for motor behaviors, **Neural computation**, 25.2, 328-373, 2013.
2. Chaumette, François, and Seth Hutchinson, **Visual servo control: I. Basic approaches**, IEEE Robotics & Automation Magazine, 13.4 82-90, 2006.
3. Chaumette, François, and Seth Hutchinson, **Visual servo control, II, Advanced approaches** [Tutorial], IEEE Robotics & Automation Magazine 14.1: 109-118, 2007.